

How Truss Bridges Carry Load

1. What a truss is

A truss is a frame of straight members joined at their ends to form triangles. A triangle cannot change shape unless one of its sides changes length, so a frame of triangles stays rigid under load. A rectangle, by contrast, can lean into a parallelogram without any side changing length, which is why square frames need a diagonal.

2. Tension and compression

Every member of a truss is either pulled or pushed along its length. A member being pulled is in tension and tends to stretch. A member being pushed is in compression and tends to shorten or buckle. In a simple bridge truss the top chord is in compression and the bottom chord is in tension, while the diagonals alternate between the two.

3. The load path

When a vehicle crosses the bridge, its weight first presses on the deck. The deck passes the weight to the joints of the truss. The members carry the force from joint to joint toward the two ends. Finally the supports at each end push up on the truss and pass the load into the ground. This route is called the load path.

4. Why trusses use less material

A solid beam bends under load, so most of its material near the middle of its depth does little work. A truss puts material only where force flows, along the chords and diagonals. For the same span, a truss can therefore be lighter than a solid beam while carrying the same load.

5. How trusses fail

Long thin members in compression can buckle sideways before the material itself is crushed. Joints can also fail if the connection is weaker than the members it joins. Engineers therefore check each member for buckling and design joints to be stronger than members.